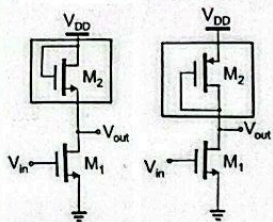


100

diff in body effect



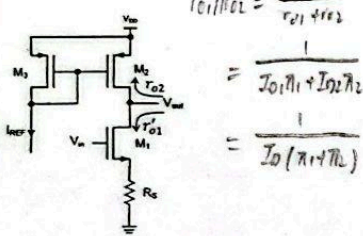
$$A_v = -g_{m1}(r_{o1} || r_{diode})$$

$$= -g_{m1}(r_{o1} || \frac{1}{g_{m2} + g_{mb2}} || r_{o2})$$

$$\approx -g_{m1} \frac{1}{g_{m2} + g_{mb2}} = -\frac{g_{m1}}{g_{m2}} \frac{1}{1 + \eta}$$

$$= -\sqrt{\frac{(W/L)_1}{(W/L)_2}} \frac{1}{1 + \eta}$$

ONLY IF BOTH ARE N/P
Top swing limited by VDD - |VTH2|



$$G_m = \frac{g_{m1} \frac{1}{R_s}}{g_{m1} + g_{mb1} + \frac{1}{R_s} + \frac{1}{r_{o1}}}$$

$$= \frac{r_{o1} g_{m1}}{r_{o1} R_s (g_{m1} + g_{mb1}) + r_{o1} + R_s}$$

If $g_{m1} + g_{mb1} \gg \frac{1}{R_s} \& \frac{1}{r_{o1}}$,

$$G_m \approx \frac{1}{(1 + \eta) R_s}$$

$$r_{o1}' = [1 + \frac{R_s}{r_{o1}} + (g_{m1} + g_{mb1}) R_s] r_{o1}$$

$$r_{o1}' \approx [1 + (g_{m1} + g_{mb1}) R_s] r_{o1}$$

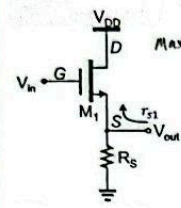
$$A_v = G_m r_{o1}'$$

$$= \frac{-r_{o1}' g_{m1}}{r_{o1}' R_s (g_{m1} + g_{mb1}) + r_{o1}' + R_s}$$

$$\approx \frac{-r_{o2}}{(1 + \eta) R_s} (\approx -R_{top}/R_{bot})$$

• use exact first

(swing) = (V_{in} - V_{TH}) - V_{DS} bottom



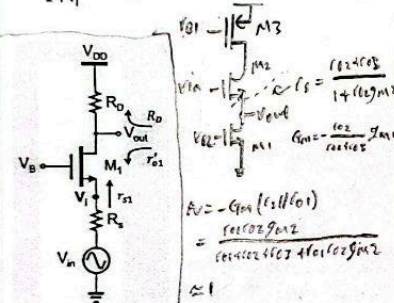
$$r_{s1} || R_s = \frac{1}{g_{m1} + g_{mb1} + \frac{1}{r_{o1}}} || R_s$$

$$= \frac{1}{g_{m1} + g_{mb1} + \frac{1}{R_s} + \frac{1}{r_{o1}}}$$

$$A_v = \frac{g_{m1}}{g_{m1} + g_{mb1} + \frac{1}{R_s} + \frac{1}{r_{o1}}}$$

$$\approx \frac{g_{m1}}{g_{m1} + g_{mb1}} \text{ if } g_{m1} \gg \frac{1}{R_s} + \frac{1}{r_{o1}}$$

$$= \frac{1}{1 + \eta} \approx 1 \text{ for } R_p = 0 \text{ } (g_{m1} \approx -\eta)$$



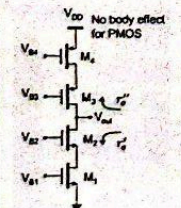
small input impedance, large output impedance, good for current amplifier

$$A_v = \frac{r_{o1}}{R_s + r_{s1}} (\frac{1}{r_{o1}} + g_{m1} + g_{mb1}) (r_{o1}' || R_D)$$

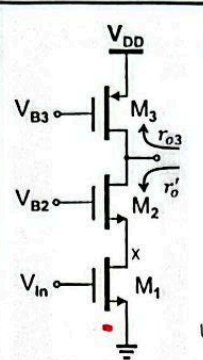
$$\approx \frac{r_{s1}}{R_s + r_{s1}} (g_{m1} + g_{mb1}) (r_{o1}' || R_D)$$

$$r_{s1} = \frac{r_{o1} + R_D}{1 + (g_{m1} + g_{mb1}) r_{o1}} \approx \frac{1}{g_{m1} + g_{mb1}}$$

$$r_{s1} = \frac{r_{o1} + r_{o2}}{1 + (g_{m1} + g_{mb1}) r_{o1}} \approx \frac{2}{g_{m1} + g_{mb1}}$$

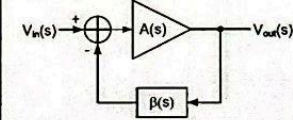


top headroom
V_{DD3} + V_{DD4}
bottom headroom
V_{DD1} + V_{DD2}



$A_v = -G_m (r_{o1}' || r_{o3})$
 $A_v \approx -g_{m1} r_{o3}$
 $G_m = \frac{(g_{m2} + g_{mb2} + \frac{1}{r_{o2}}) r_{o1} g_{m1}}{1 + (g_{m2} + g_{mb2} + \frac{1}{r_{o2}}) r_{o1}} \approx g_{m1} \text{ CSL}$
no current gain from CG
 $r_{o1}' = [1 + \frac{r_{o1}}{r_{o2}} + (g_{m2} + g_{mb2}) r_{o1}] r_{o2}$
 $r_{o1}' \approx [1 + (g_{m2} + g_{mb2}) r_{o1}] r_{o2}$
CGD1 is not Millered and amplified to be very big at the input, lower input cap. High BW

General closed-loop amplifier:



A(s) - Open-loop transfer function (or gain)
beta(s) - Feedback path transfer function
A(s)beta(s) - Loop gain

Output = "direct" / (1 + "loop") * Input

Note that its feedback to -ve summer

Closed-loop gain:

$$v_{out}(s) = A(s)[v_{in}(s) - \beta(s)v_{out}(s)]$$

$$\Rightarrow \frac{v_{out}(s)}{v_{in}(s)} = \frac{A(s)}{1 + A(s)\beta(s)}$$

If A(s)beta(s) >> 1, The closed-loop gain becomes

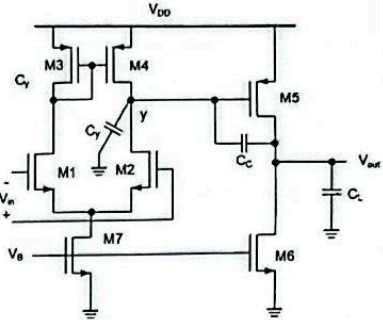
$$\frac{v_{out}(s)}{v_{in}(s)} \approx \frac{1}{\beta(s)}$$

$$I_1 = I_2 \left(\frac{1/R_1}{1/R_1 + 1/R_2 + \dots} \right)$$

can ignore

if V_{ic} not given; SW from V_{DD} - V_{TH1} to V_{DD2} + V_{DD7}

if V_{ic} given; SW from V_{DD} - V_{TH4} to V_{ic} - V_{TH1}



Miller op-amp with D to S + CS
AC is just stack
DC: input common mode voltage is limited by tail voltage (saturation of current mirror) and amplifier mosfet VTH
too much current flow leading to amplifier MOSFET saturation (active CSL is always in saturation)
offset: find the voltage for output CS to give midpoint, then output of D to S need to give the VGS required. Check current mirror and active CSL mirror

Simplified before comp

$$C_1 = C_y + (1 + g_{m5}(r_{o5} || r_{o6})) C_{gd5}$$

$$C_2 = C_L + C_{gd5}$$

$$\omega_{p1} = \frac{1}{(r_{o2} || r_{o4}) C_1} \quad \omega_{p2} = \frac{1}{(r_{o5} || r_{o6}) C_2}$$

$$\omega_{z1, RHP} = \frac{g_{m5}}{C_{gd5}} \quad f = \frac{g_{m5}}{2\pi C_{gd5}}$$

After C_c ≈ PF

$$LHP \omega_{p1} \approx \frac{1}{r_{out1}(1 + g_{m5} r_{out2}) C_c}$$

$$LHP \omega_{p2} \approx \frac{g_{m5}}{C_y (\frac{C_c}{C_1} + 1) + C_1} \approx \frac{g_{m5}}{C_1} \propto \frac{1}{L + C_c}$$

$$RHP \omega_{z1} = \frac{g_{m5}}{C_c}$$

Adding R_c

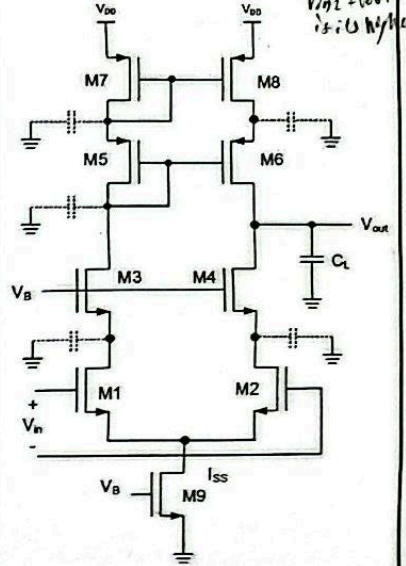
$$\omega_{z2} = \frac{1}{C_c (1/g_{m5} - R_c)}$$

$$\omega_{p2} \rightarrow \frac{-g_{m5} C_c}{C_c C_c + C_c C_c + C_c C_c} = \frac{C_c (1/g_{m5} - R_c)}{C_c (1/g_{m5} - R_c)}$$

$$R_c = \frac{1}{g_{m1}} \left(1 + \frac{C_c C_c / C_c + C_c C_c + C_c}{C_c} \right) \approx \frac{1}{g_{m1}} \left(\frac{C_c + C_c}{C_c} \right)$$

If C_c >> C₁, C₂

$$\omega_u = A_v \omega_{3dB} = \frac{g_{m2}}{C_c}$$



Telescopic op-amp
high output resistance, for current output
D to D: swing limited by all the VOD
D to S: swing limited by all the VOD for the amplifier and current mirror and the cascode current mirror required minimum drop of VTH + 2VOD

$$r_{out} \approx (g_{m6} r_{o6} r_{o8}) || (g_{m4} r_{o4} r_{o2})$$

$$\omega_{3dB} = \omega_{p1} \approx \frac{1}{r_{out} C_L}$$

$$\omega_{p,i} \approx \frac{g_{m,i}}{C_i} \leftarrow \text{very high freq}$$

$$\omega_u = A_v \omega_{p1} = \frac{g_{m2} r_{out}}{r_{out} C_L} = \frac{g_{m2}}{C_L}$$

can increase more current for increasing BW
Suitable to drive capacitive load unlike Miller Opamp
• DO NOT FORGET zero when millering & plotting bode
• Divide connectal use > VTH

